

SUVAT graphs, intervals and vertical motion

Recall/jargon companion for the approved 18 July lesson. Start with the prompt, say the method aloud, then check the concise outcome.

Evidence boundary: Controlled lesson content runs through 00:31:44. The only current homework worked here is Q13(a)-(b); the third selected item is unknown. No previous-lesson or projectile exercise is reconstructed.

30-second method recall

1. SUVAT requires constant acceleration in one dimension.
2. Choose a positive direction and list signs before selecting an equation.
3. For a time interval, use endpoint subtraction.
4. The n th second is $n-1$ to n .
5. For time above a height, check two physical crossings and subtract later minus earlier.

Question-method recall

Target	Source/derived status	Check answer
Q01 · Proportional graph in free fall	Officially controlled question and answer.	A: v^2 against h is a straight line through the origin; v against h is a square-root curve.
Q02 · Distance between two times	Officially controlled answer; full arithmetic is a derived calculation from the controlled values.	$0.24525 \text{ m} \approx 0.25 \text{ m}$, option B.
Q03 · Second second to fourth second	Officially controlled question and answer.	28 m, option A.
Q04 · General time above a height	Lesson/worksheet material; source and marks are unknown. The compact result is labelled as a derived extension.	Derived extension: time above $h_0 = 2\sqrt{(u^2 - 2gh_0)}/g$, provided two relevant physical crossings exist.
Q05 · Vertical projection: height, top time, time above 3 m	Lesson/worksheet material; source and marks are unknown. Full-precision Q05(c) is derived, while board rounding is retained separately.	(a) Height $5.09684... \text{ m} \approx 5.1 \text{ m}$. (b) Top time $1.01937... \text{ s} \approx 1.02 \text{ s}$. (c) Above 3 m: board rounding about 1.3 s; derived $1.30765... \text{ s} \approx 1.31 \text{ s}$.
Q13(a) · Braking-distance scaling	Officially controlled Pearson homework item.	$70(25/15)^2 = 194.444... \text{ m}$; official result 194 m, with 195 m accepted; hence about 190 m.
Q13(b) · Road-train length	Officially controlled Pearson homework item.	$54.596... \text{ m} \approx 54.6 \text{ m}$; official result 55 m.

Jargon you should be able to explain

SUVAT

MEANING

A set of equations for one-dimensional motion with constant acceleration.

AVOID

Do not use them unchanged when acceleration varies.

Displacement

MEANING

Signed change in position, measured in metres.

AVOID

It is not always the same as total distance.

Direct proportionality

MEANING

A relationship $y=kx$; its graph is a straight line through the origin.

AVOID

A straight line with non-zero intercept is not direct proportion.

Endpoint subtraction

MEANING

For an interval, calculate $s(t_2) - s(t_1)$.

AVOID

It is not merely putting Δt into a formula with the original initial velocity.

nth second

MEANING

The interval from $t=n-1$ to $t=n$.

AVOID

The fourth second is not the whole interval from $t=0$ to $t=4$.

Positive direction

MEANING

A stated choice that fixes signs. With up positive, $a = -g$.

AVOID

Gravity does not become zero at maximum height.

Crossing times

MEANING

The two times a projectile reaches a chosen height.

AVOID

Labels t_1 and t_2 are arbitrary; use later minus earlier.

Maximum height

MEANING

For vertical projection, $h_{\max} = u^2 / (2g)$ under the stated model.

AVOID

A height target is not a given value to substitute.

Braking-distance scaling

MEANING

At fixed deceleration magnitude, braking distance is proportional to initial speed squared.

AVOID

It is not proportional to speed itself.

Unit conversion

MEANING

Convert km h^{-1} to m s^{-1} by dividing by 3.6.

AVOID

Do not square a km h^{-1} value before converting.

Final checking routine

- Does the equation exclude the variable I do not need?
 - Have I answered interval distance rather than cumulative displacement?
 - Are all speeds in m s^{-1} before squaring?
 - For vertical motion, have I used one sign convention throughout?
 - Have I kept precision until the final rounded answer with units?
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